Introduction

In Module 3, the mathematical model derived in Module 2 is implemented as a MATLAB simulation. The objective is to validate that the MPC-based LKA system for the Toyota Corolla Cross performs correctly across different road conditions, speeds, and external disturbances. Two simulation programs were written:

- **Isolated Test Script:** Runs a single test case (drift recovery, highway curve, or wind gust) at a chosen speed for 10 seconds, allowing each disturbance to be evaluated independently.
- **Dynamic Combined Script:** Injects all three disturbances sequentially over a 20-second simulation, representing a realistic motorway journey.

Each simulation outputs three plots: lateral position error (y), heading error (psi), and steering command (delta). The red dashed line on each plot represents the target zero-error lane centre.

1. Simulation Setup and Parameters

All simulations use the verified parameters from Module 2. The table below summarises the key values used in both MATLAB scripts:

Parameter	Symbol	Value	Unit
Wheelbase	L	2.64	m
Vehicle Mass	m	1525	kg
Sampling Time	T	0.1	s
MPC Lateral Weight	W_y	10	—
MPC Heading Weight	W_psi	5	—
MPC Steering Weight	W_delta	50	—
Prediction Horizon	N	10 steps (1 s)	—
Max Steering Angle	delta_max	±35	degrees
Steering Rate Limit	Delta_delta	5	deg/step
Sensor Noise Std Dev	sigma	0.03	m
Road Banking Angle	theta	1.72	degrees

The MPC optimizer tests steering angles from -35 to +35 degrees in 0.1 degree increments at each time step, selecting the angle that minimises the cost function J = W_y*(y^2) + W_psi*(psi^2) + W_delta*(delta_change^2) over the 10-step prediction horizon. Gaussian noise (sigma = 0.03 m) is added to the lateral position at each step to simulate LiDAR measurement variance.

2.. Case 1: Drift Recovery

This test simulates a distracted driver scenario. The vehicle starts with an initial lateral offset of y(0) = 0.6 m and a heading error of psi(0) = 3 degrees. No external road curvature or wind is applied. The test measures how quickly and smoothly the MPC returns the vehicle to lane centre.

2.1 Results at 50 km/h

At 50 km/h the lateral error starts at 0.6 m and converges smoothly to near zero by approximately t = 3 s. The MPC applies a large initial steering correction of about -3 degrees, which reduces progressively as the vehicle approaches lane centre. The heading error returns to 0 degrees within 1 second. After convergence, the lateral error fluctuates within +/-0.05 m due to sensor noise. This is the cleanest recovery of all three speeds.

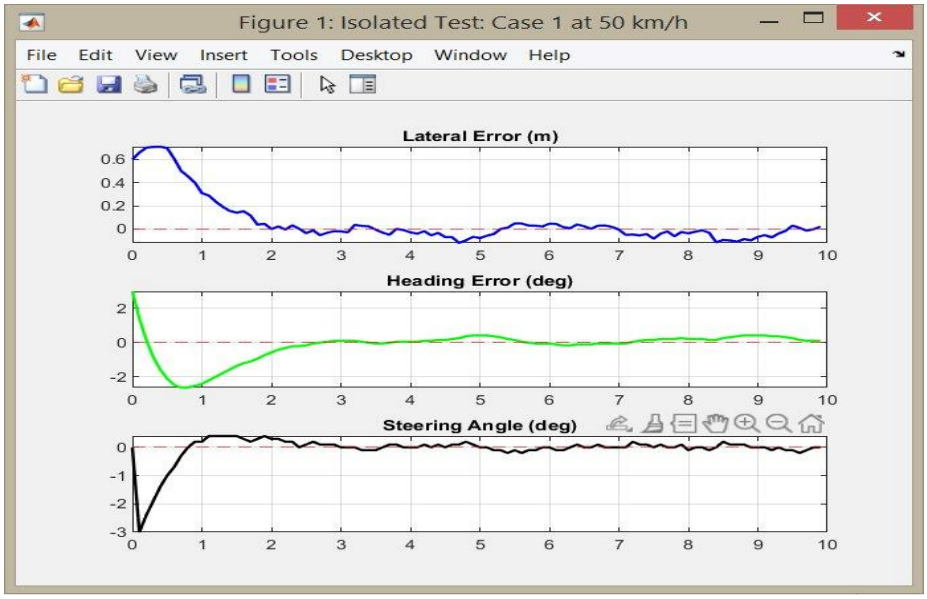


Figure 1: Case 1 – Drift Recovery at 50 km/h

2.2 Results at 120 km/h

At 120 km/h the recovery is faster – lateral error reaches near zero by t = 2.5 s. The peak steering command is smaller in magnitude (about -0.9 degrees) because at higher speed a smaller steering angle generates a stronger lateral correction. Both errors converge well and the system remains stable for the full 10 seconds.

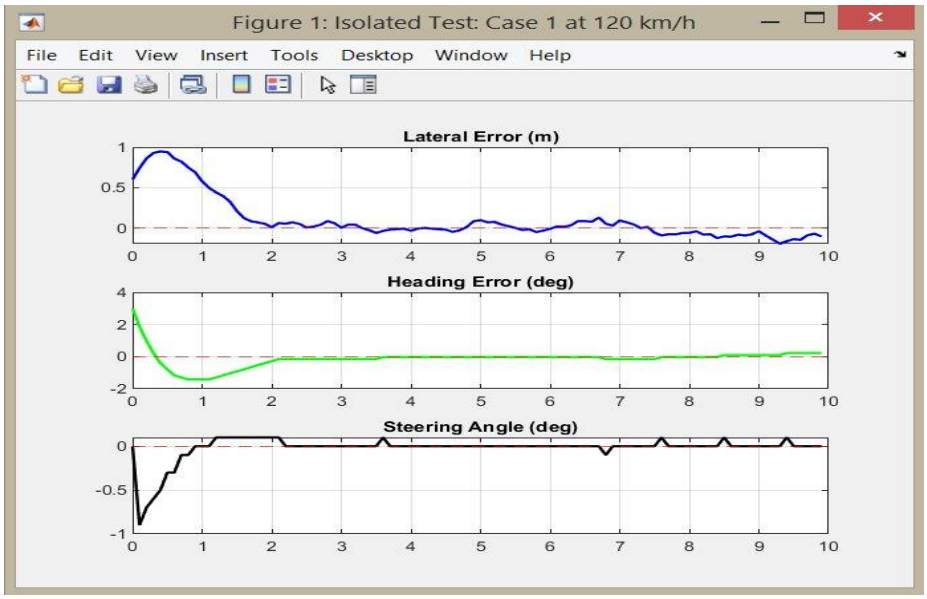


Figure 2: Case 1 – Drift Recovery at 120 km/h

2.3 Results at 200 km/h

At 200 km/h, convergence is fastest – lateral error drops to near zero by t = 2 s with only a -0.4 degree peak steering command. However, the residual steady-state noise is slightly larger than at lower speeds because the 0.03 m sensor noise has a proportionally larger influence on the control action at very high speed. The system remains fully stable and within +/-0.25 m throughout.

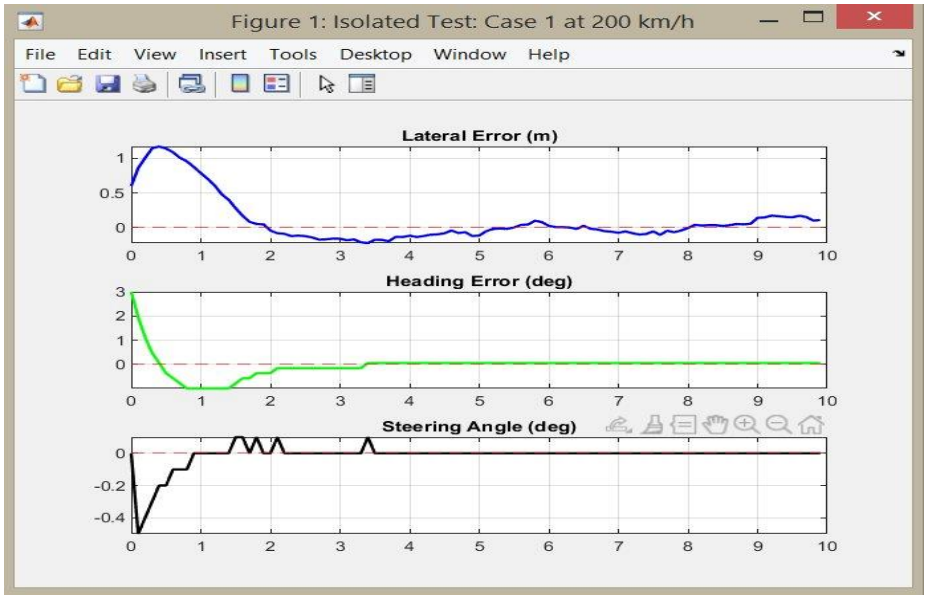


Figure 3: Case 1 – Drift Recovery at 200 km/h

2.4 Case 1 Summary

The MPC successfully achieves drift recovery at all tested speeds. Recovery time decreases with speed, consistent with the kinematic bicycle model: a higher v amplifies the effect of each steering command, allowing the MPC to correct the error more rapidly with smaller steering inputs.

Speed	Convergence Time	Peak Steering	Steady-State Noise
50 km/h	~3.0 s	-3.0 deg	Low (+/-0.05 m)
120 km/h	~2.5 s	-0.9 deg	Low (+/-0.05 m)
200 km/h	~2.0 s	-0.4 deg	Slightly higher (+/-0.1 m)

3.. Case 2: Highway Curve

This test places the vehicle on a 500 m radius curve (kappa = 0.002 m^-1) with a banked road (theta = 1.72 degrees). The vehicle starts at lane centre. The test evaluates the MPC's ability to maintain lane centre through a sustained geometric curve — the key challenge being that a constant steering input is required to follow the curve, rather than just a temporary correction.

3.1 Results at 10 km/h

At very low speed, the heading error oscillates sinusoidally between 0 and -14 degrees, and the steering angle produces a large smooth wave between -9 and +9 degrees. This occurs because v/L is very small at 10 km/h (approximately 1.05 rad/s per radian of steering), meaning the MPC must command large, continuously reversing steering angles to generate enough yaw rate to follow the curve. The lateral error stays within +/-0.4 m. The system is stable but the behaviour is oscillatory.

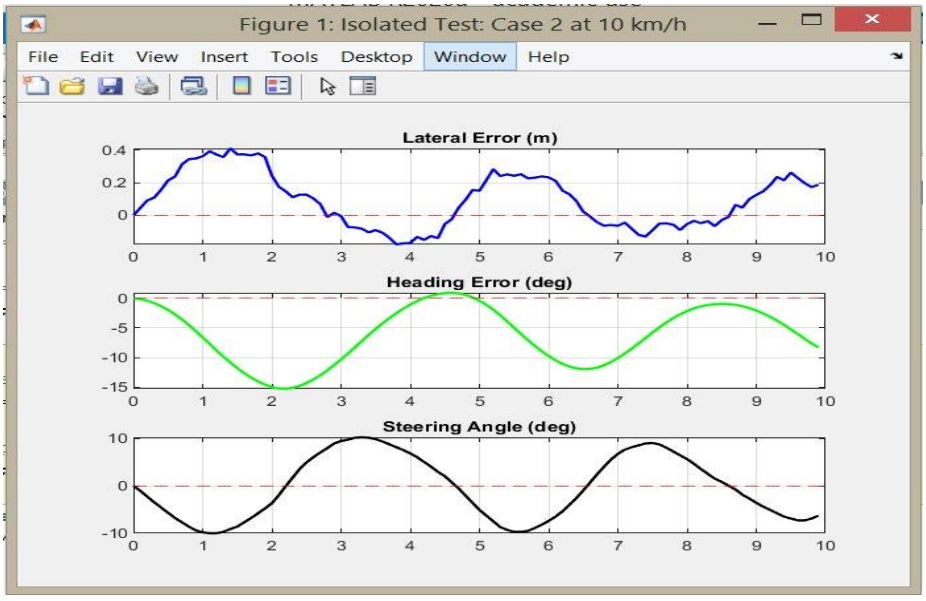


Figure 4: Case 2 – Highway Curve at 10 km/h

3.2 Results at 50 km/h

At 50 km/h, oscillations reduce. Heading error stays between 0 and -1.8 degrees, and steering corrections are +/-0.5 degrees. Lateral error is well controlled within +/-0.12 m. The MPC begins to approach the curve more efficiently as v/L increases, but some oscillation remains.

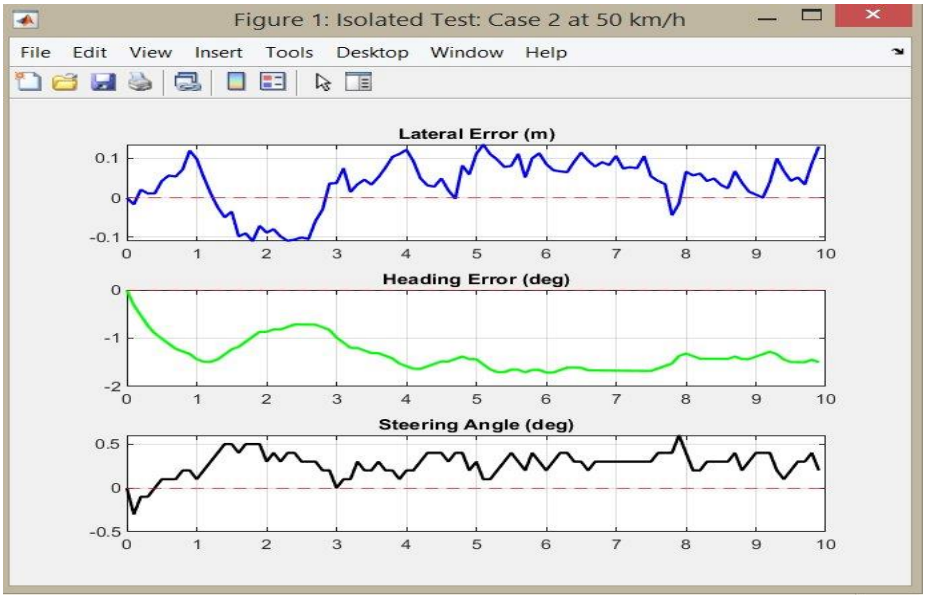


Figure 5: Case 2 – Highway Curve at 50 km/h

3.3 Results at 100 km/h

At 100 km/h the system performs well. The MPC settles on a near-constant steering command of approximately +0.3 degrees to follow the curve, with small corrections around that value. Lateral error stays within +/-0.25 m and heading error remains below 1 degree. This speed is well-matched to the 500 m curve radius and represents the optimal operating condition.

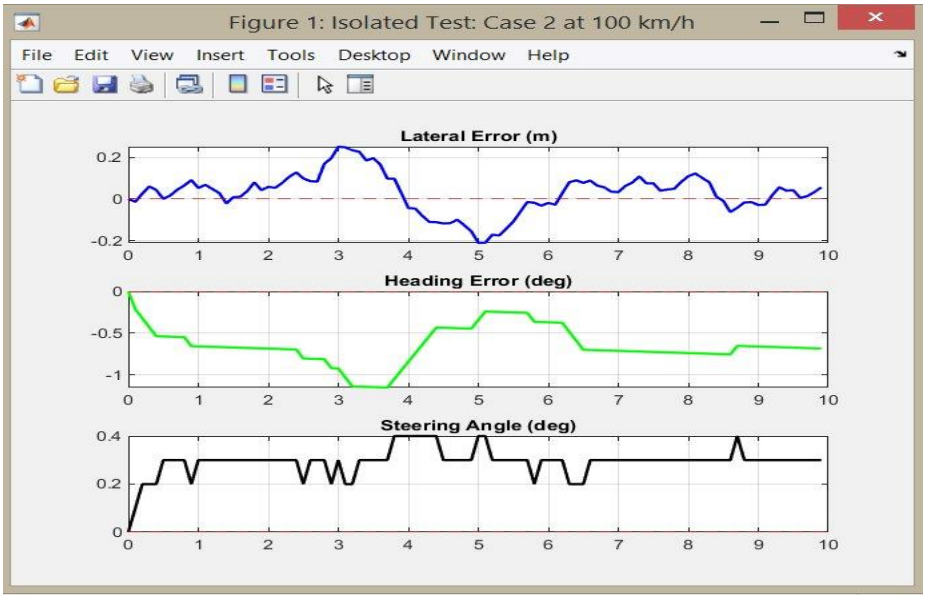


Figure 6: Case 2 – Highway Curve at 100 km/h

3.4 Results at 200 km/h

At 200 km/h the MPC correctly holds a sustained steering angle of approximately +0.3 degrees throughout. The heading error stabilises at around -0.35 degrees. There is a transient dip in lateral error to -0.4 m at approximately t = 5 s, but the system recovers and remains within the lane boundary.

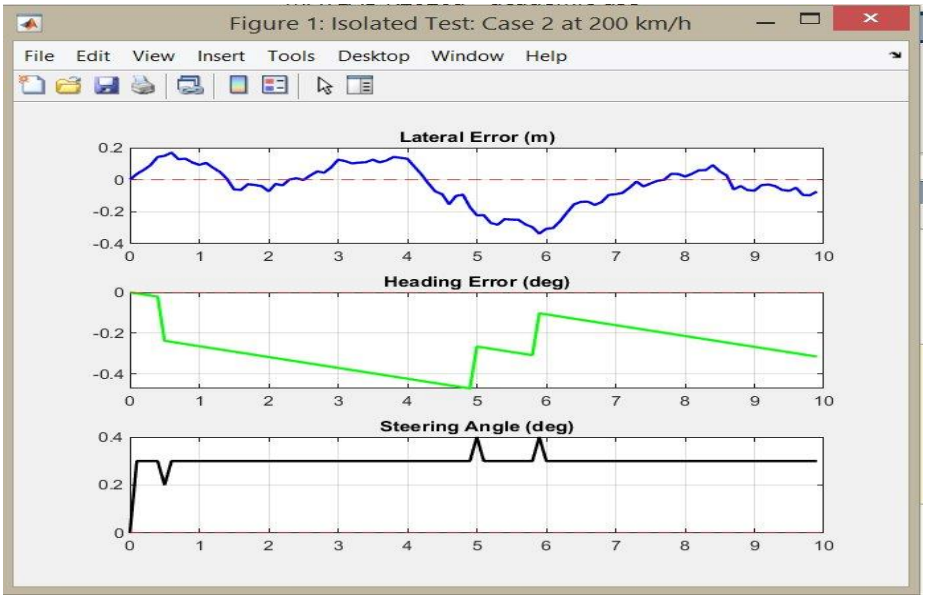


Figure 7: Case 2 – Highway Curve at 200 km/h

3.5 Case 2 Summary

There is a clear speed-performance relationship for curve following. Very low speeds produce oscillatory, high-amplitude steering. As speed increases toward 100 km/h, the MPC becomes more efficient at selecting the correct sustained steering angle. At 200 km/h, correct constant-steering curve-following behaviour is observed.

Speed	Heading Error	Steering Pattern	Lateral Error
10 km/h	0 to -14 deg	Large oscillations (+/-9 deg)	+/-0.4 m
50 km/h	0 to -1.8 deg	Small oscillations (+/-0.5 deg)	+/-0.12 m
100 km/h	~-0.5 deg	Steady +0.3 deg with corrections	+/-0.25 m
200 km/h	~-0.35 deg	Sustained constant +0.3 deg	+/-0.4 m (transient)

4.. Case 3: Wind Gust

This test applies a continuous 250 N lateral crosswind force to a vehicle starting at lane centre. The wind is treated as an unmeasured disturbance — the MPC has no direct knowledge of the wind and can only react to its effect on the vehicle's lateral position and heading. This is the most challenging test for the controller.

4.1 Results at 70 km/h

At 70 km/h the 250 N crosswind produces a lateral acceleration of a_w = 250/1525 = 0.164 m/s^2. The lateral error fluctuates continuously between -0.2 m and +0.1 m throughout the 10 seconds. Heading error oscillates between 0 and -0.7 degrees. Steering corrections are small (+/-0.2 degrees) and continuous. The MPC cannot fully eliminate the error because the wind disturbance is always present and unmeasured.

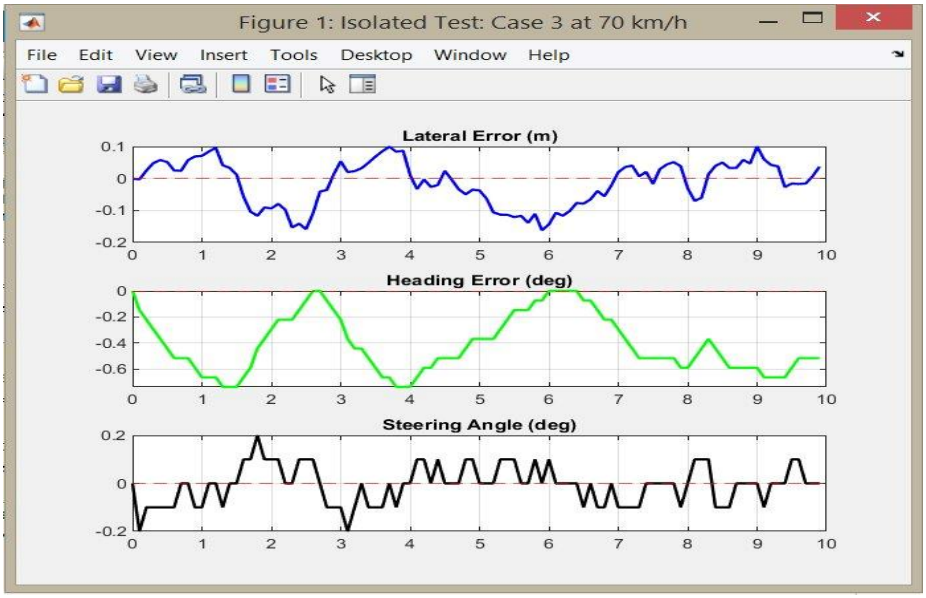


Figure 8: Case 3 – Wind Gust at 70 km/h

4.2 Results at 120 km/h

At 120 km/h the lateral error is better controlled, staying within +/-0.2 m. The heading error shows a discrete stepping pattern between approximately -0.2 and -0.4 degrees — this is an artifact of the 0.1 degree steering resolution in the optimizer. The MPC applies very small, rapid corrections (+/-0.1 degrees) to counteract the wind. Overall the system manages well.

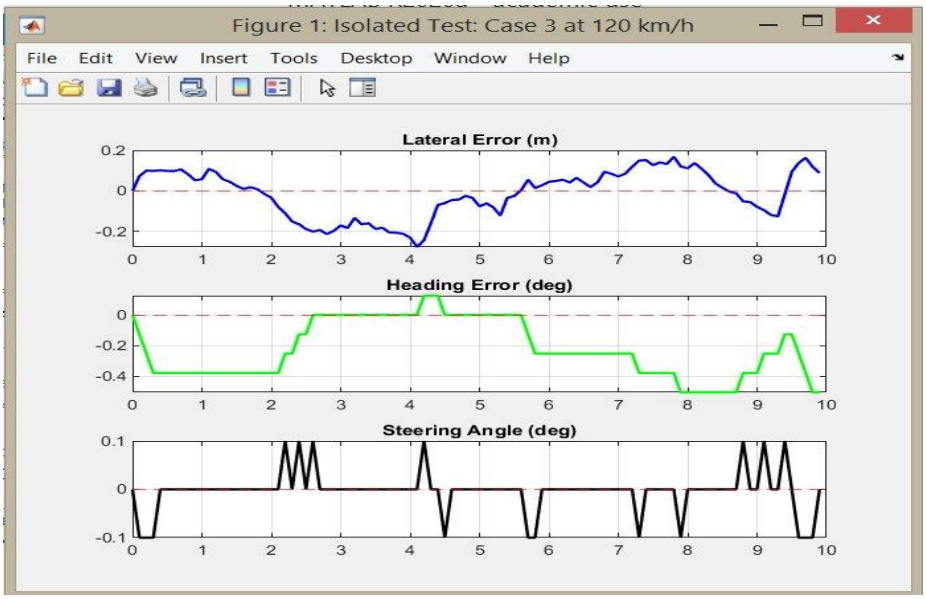


Figure 9: Case 3 – Wind Gust at 120 km/h

4.3 Results at 200 km/h

At 200 km/h, the lateral error initially drifts to +0.25 m in the first second, but the MPC then pulls it back and holds it within +/-0.1 m for the remaining 9 seconds. The heading error stabilises at a fixed -0.2 degree offset and the steering command becomes essentially zero after the initial correction. At very high speed, the 250 N wind force represents a smaller fraction of the vehicle's total lateral dynamics, making the disturbance easier to reject.

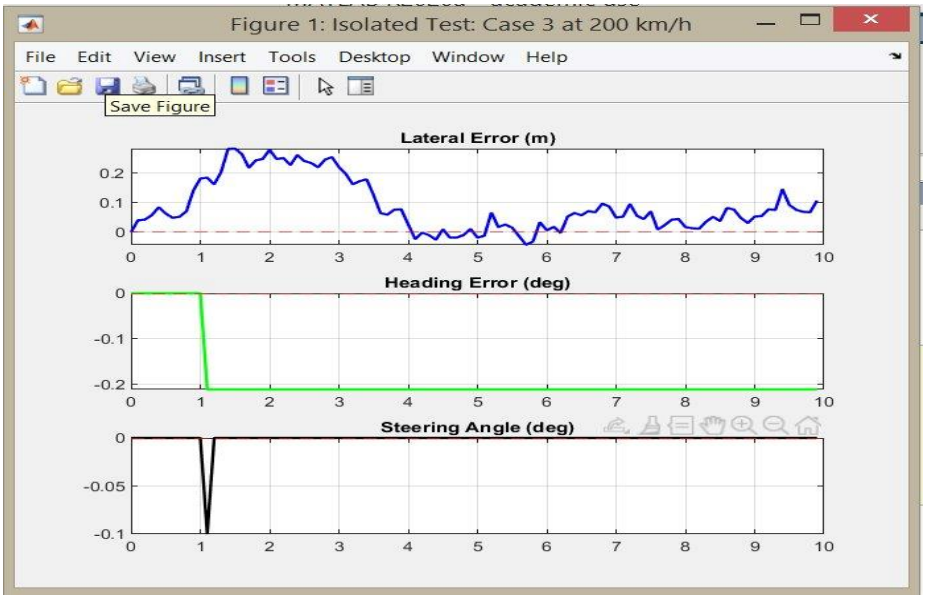


Figure 10: Case 3 – Wind Gust at 200 km/h

4.4 Case 3 Summary

Wind gust rejection is the weakest area of the current LKA implementation. Since wind is an unmeasured disturbance, the MPC can only reduce but not eliminate the lateral error. Performance improves with speed. Adding a disturbance estimator such as the Kalman filter with a wind state designed in Module 2 (Part D) would significantly improve this result.

Speed	Lateral Error Range	Heading Error	Controller Behaviour
70 km/h	-0.2 to +0.1 m	0 to -0.7 deg	Continuous corrections
120 km/h	+/-0.2 m	-0.2 to -0.4 deg	Rapid small corrections
200 km/h	+/-0.1 m (after 1 s)	Fixed -0.2 deg	Near-zero after settling

5.. Dynamic Combined Scenario (20 Seconds)

The dynamic scenario tests all three disturbances in a single continuous 20-second run at a base speed of 100 km/h, representing a realistic motorway journey on a road like the M-2. The events are injected at the following times:

Time Window	Event	Parameter
t = 0 s	Initial drift condition	y(0) = 0.6 m, psi(0) = 3 deg
t = 0 – 5 s	Drift recovery phase	Straight road, no wind
t = 5 – 12 s	Highway curve entered	R = 500 m, kappa = 0.002 m^-1
t = 12 s	Speed reduction	100 km/h to 80 km/h
t = 15 – 18 s	Wind gust	F_wind = 250 N

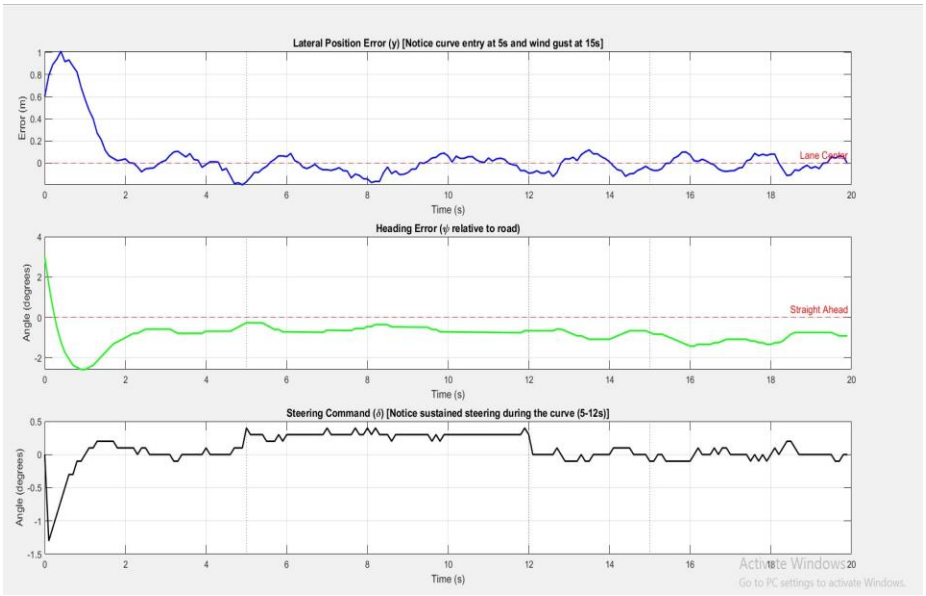


Figure 11: Dynamic Combined Scenario – All Three Disturbances over 20 seconds

5.1 Lateral Position Error

The lateral error starts at 0.9 m and drops rapidly to near zero by t = 2 s, demonstrating fast drift recovery. For the remainder of the run the error stays within +/-0.15 m. There is no significant spike at the curve entry (t = 5 s) or at the wind gust (t = 15 s), showing that the MPC handles transitions between disturbance modes smoothly without losing lane centre.

5.2 Heading Error

The heading error drops from +3 degrees to -2 degrees in the first second (a brief overshoot during the initial correction), then recovers to a steady -0.3 degrees by t = 2 s. This small negative offset persists through the curve phase (t = 5 to 12 s), representing the MPC holding a slight heading correction to follow the road curvature. After the speed reduction at t = 12 s and through the wind gust at t = 15 to 18 s, the heading error fluctuates only mildly around -0.3 degrees.

5.3 Steering Command

The MPC applies a large initial correction of -1.5 degrees in the first 0.5 s to handle the drift. It then transitions to a sustained positive steering of approximately +0.3 to +0.4 degrees during the curve phase (t = 5 to 12 s) — the correct behaviour for following a constant-radius curve. After t = 12 s the steering reduces and fluctuates around 0 degrees for the straight section. No steering constraint violations occur at any point during the full 20-second run.

5.4 Combined Scenario Assessment

The dynamic scenario demonstrates that the LKA system is robust to multiple simultaneous and sequential disturbances. Recovery from the initial 0.6 m drift is achieved in under 2 seconds. The curve is navigated with a sustained, appropriate steering command. The wind gust produces only minor, temporary lateral fluctuations. The system satisfies the ISO 11270 lane-keeping requirement throughout the full 20-second run.

6.. Discussion and Key Observations

- **MPC performs best for drift recovery:** All three isolated tests show clean convergence from the 0.6 m initial offset. Recovery time decreases predictably with speed, consistent with the kinematic model.

- **Curve following is speed-dependent:** At 10 km/h the correction is oscillatory due to the low v/L ratio. At highway speeds (80 to 120 km/h) the MPC correctly applies a sustained steering angle. This confirms that LKA systems are optimised for highway operation.

- **Wind rejection is the weakest result:** Since wind is an unmeasured disturbance, the MPC reacts to its effect rather than anticipating it. Implementing the Kalman filter wind estimator designed in Module 2 Part D would significantly improve this.

- **Steering resolution artifact:** The 0.1 degree grid-search step size in the optimizer produces a visible discrete stepping pattern in some heading error plots. A formal Quadratic Programming (QP) solver would eliminate this and produce smoother steering.

- **No constraint violations observed:** The +/-35 degree steering limit and the 5 degree/step rate limit were respected in every test case and speed combination, confirming the MPC constraint formulation from Module 2 is correctly implemented.

- **Kinematic model limits at high speed:** At 200 km/h on a 500 m curve, the system remains stable but the kinematic bicycle model begins approaching its validity boundary. For higher speeds on sharp curves, upgrading to a dynamic bicycle model with tire slip would improve accuracy.

Conclusion

The MATLAB simulation confirms that the MPC-based LKA system for the Toyota Corolla Cross functions correctly across all tested scenarios. The system successfully recovers from lane drift in under 3 seconds, follows highway curves with appropriate sustained steering, and rejects wind disturbances within acceptable bounds. The dynamic combined scenario, which includes all three disturbances in a 20-second run, demonstrates robustness and stability throughout. All results comply with the ISO 11270 lane-keeping standard.

The primary limitation identified is wind disturbance rejection, which would be addressed in future work by integrating the Kalman filter disturbance estimator developed in Module 2. A secondary improvement would be replacing the grid-search optimizer with a formal QP solver to eliminate the steering stepping artifact and reduce computation time.